SESSION 04 / PRINTABLE GUIDE

Independent challenge: Explain a device's daily readings

Session 04 | Read-only exit exercise
Output: A small, reproducible analysis and an evidence-backed explanation, without step-by-step transformation code.

Complete Notebook 101 and the successful main scenario notebook first. Use the same approved default Lakehouse and batch `s04-lab-devices-001`. Warehouse publication is not required for this Lakehouse analysis and must not be inferred from its success.

Import the challenge starter. It supplies context/table reads, two code-task placeholders, and a Markdown answer area. Replace the labeled `NotImplementedError` stops with your analysis. Do not open the presenter solution until the learner has submitted an attempt or requested recovery.

The business question

Operations asks:

> For **D032 on October 10, 2026 (UTC)**, what were the accepted reading count, minimum, maximum, and average temperature? How does the reported daily average compare with **October 6**? Were all scheduled observations accepted, and what can we legitimately conclude about device health?

These are different device/date selections from the guided coverage checkpoint. The source contract schedules one observation every five minutes: **288 per device/day**. That is a known teaching denominator, not something inferred from the number of records that arrived.

Constraints

- Read existing `s04_lab_device_silver`, `s04_lab_device_bronze`, and `s04_lab_device_rejected`, filtered to the intended batch.
- Use typed **accepted Silver events** for temperature metrics; do not re-clean Bronze or quietly change the acceptance policy.
- Preserve the device/day grain and useful site/type context. Filter to the two requested UTC dates before presenting results.
- Do not write, append, overwrite, publish, install packages, change permissions, or edit source files. Keep driver-side output small.

- Use the survival guide and PySpark reference as lookup aids, not a copied worked transformation.

Task 1: Produce the daily summary

Return one row per requested device/day with:

Required field	Meaning
UTC date, device, site, device type	Scope and grain
Accepted reading count	Not the raw source count
Minimum, maximum, average temperature	Accepted events only; show the mean to two decimal places
Scheduled readings and accepted-slot shortfall	288 and its difference from accepted count
Accepted percentage	Accepted count / scheduled readings, labeled clearly

Compare the two **reported rounded daily means** without calling the difference statistically significant or treating their average as an overall event mean.

Task 2: Explain the source and quarantine evidence

Count raw source rows and primary quarantine reasons for D032 from each of the two daily **source exports**. Include a zero-rejection day in your explanation, even if the grouped quarantine result has no row for it.

Use `source_file` to identify the relevant `readings_YYYY-MM-DD.csv` export. A record with an invalid timestamp cannot reliably be assigned an event day by parsing that timestamp; an export filename supplies source-scope evidence, not a repaired business timestamp.

Explain why raw row count, quarantine row count, accepted count, and scheduled slots are different populations. Both occurrences of an ambiguous duplicated event were rejected under the existing policy.

Task 3: Write the interpretation

In the notebook's Markdown answer area, state:

1. The input tables, batch, device, dates, and output grain.
2. The two summaries and the direction/magnitude of the reported-mean change.
3. Whether accepted coverage is complete on the requested date and the evidence.
4. What zero quarantine rows does **not** prove, and why shortfall is not automatically device downtime.

5. Why two daily summaries do not establish device health, a trend, statistical significance, or a firmware-related cause.
6. Where the computation ran, whether anything was persisted, and which outputs this exercise does not prove (for example Warehouse Gold/model freshness).

Add a short Markdown title/question and scope above your analysis. Use autocomplete/API help when useful. Preserve any error and explain its correction rather than suppressing it.

Completion rubric

Evidence	Accept when
Scope/context	Correct default Lakehouse, batch, device, two UTC dates, and existing approved tables
Computation	Both device/day rows, correct accepted-only metrics, explicit denominator and no unintended grain change
Quality reasoning	Raw/export/quarantine/accepted populations distinguished, including an explicit zero-rejection case
Interpretation	Numerical comparison and honest completeness/ causality limitations
Reproducibility/safety	Fresh-session run succeeds; no writes, secrets, unbounded collections, or bypassed controls

The presenter compares results with the separate solution/answer key and checks the learner's explanation, not just cell status. Restart your own session and rerun the completed challenge from the top; the setup reads persisted main-lab tables, so it must not depend on variables left in another notebook.

Presenter: Rehearse this after the main notebook and record actual Fabric results in the live-workspace test sequence. Offline checks do not prove tenant import, execution, access, or UI behavior.

Source: 10_independent_challenge.md. Companion-material links open the repository originals.